

Mark schemes

Q1.

- (a) (i) Reflection

Reflection stated for M1

M1

In (the plane) $z = 0$ (or in the $x - y$ plane)

Either version for A1

A1

2

- (ii) Rotation

Rotation stated.

M1

About the y -axis

y -axis

A1

through $\frac{\pi}{3}$ radians
(or 60°)

B1

3

(b) $T_1 T_2 = \begin{bmatrix} \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \\ 0 & 1 & 0 \\ -\frac{\sqrt{3}}{2} & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 & -\frac{\sqrt{3}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{3}}{2} & 0 & -\frac{1}{2} \end{bmatrix}$

M1 correct order of matrices

A1 fully correct

$[N.B. \begin{bmatrix} \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{3}}{2} & 0 & -\frac{1}{2} \end{bmatrix} \text{ scores } M0A0]$

M1A1

2

- (c) (i) For line of invariants points

$$\begin{bmatrix} k & 2 & -1 \\ 1 & 1 & 1 \\ 3 & 4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Set up equations – uses $M \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

M1

$$\Rightarrow kx + 2y - z = x \Rightarrow (k - 1)x + 2y - z = 0 \text{ ①}$$

$$x + y + z = y \Rightarrow x + z = 0 \text{ ②}$$

Two equations correct

A1

$$3x + 4y + z = z \Rightarrow 3x + 4y = 0 \text{ ③}$$

All three equations correct

A1

From ② $z = -x$

From ③ $y = \frac{-3x}{4}$

Defines variables in terms of one letter or 2 components in

$$\begin{bmatrix} 4 \\ -3 \\ -4 \end{bmatrix} \text{ correct}$$

M1

Substitute in ① $(k - 1)x - \frac{3}{2}x + x = 0$

$$x \left[k - 1 - \frac{3}{2} + 1 \right] = 0$$

$$x \left[k - \frac{3}{2} \right] = 0$$

Substitution into other equations or $\begin{bmatrix} 4 \\ -3 \\ -4 \end{bmatrix}$ correct

A1

$$x \neq 0 \Rightarrow k = \frac{3}{2}$$

k -value obtained.

A1

(ii) Line $x = \frac{-4}{3} y = -z$

$$\frac{x}{4} = \frac{y}{-3} = \frac{z}{-4}$$

Or equivalent

B1cao

7

Alternative:

$$\begin{bmatrix} k-1 & 2 & -1 & 0 \\ 1 & 0 & 1 & 0 \\ 3 & 4 & 0 & 0 \end{bmatrix}$$

Substitute and set up

(M1)

$$r_1 \rightarrow r_1 + r_2$$

Row operation

(A1)

$$\begin{bmatrix} k & 2 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 3 & 4 & 0 & 0 \end{bmatrix}$$

$$r_1 \rightarrow r_1 - \frac{1}{2} r_3$$

Row operation

(A1)

$$\begin{bmatrix} k - \frac{3}{2} & 2 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 3 & 4 & 0 & 0 \end{bmatrix} \Rightarrow k = \frac{3}{2}$$

k-value obtained

(A1)

$$v = \lambda \begin{pmatrix} 4 \\ -3 \\ -4 \end{pmatrix}$$

M1 obtains v in terms of single vector.

A1 correct

(M1A1)

$$\frac{x}{4} = \frac{y}{-3} = \frac{z}{-4}$$

Correct form

(B1cao)

(7)

[14]

Q2.

(a)

$$\left. \begin{array}{l} \text{direction ratios of line} = p : 3 : -1 \\ \text{normal to plane} = 1 : 1 : 2 \end{array} \right\} \begin{array}{l} \text{not} \\ \text{equal} \end{array}$$

Accept not parallel or showing vector product is non zero

B1

1

(b) $x = 3 + pt$
 $y = q + 3t$
 $z = 1 - t$

Parametric form seen

M1

Meets plane $\Rightarrow 3 + pt + q + 3t + 2(1 - t) = 10$

$\Rightarrow (5 + q) + t(p + 1) = 10$

Correct substitution in plane

A1

Within plane $\Rightarrow q = 5, p = -1$

M1 – Finding one correct value

A1 – Both values correct

M1A1

4

Alternative:

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 Point $(3, q, 1)$ is common to line and plane

Hence $\begin{pmatrix} 3 \\ q \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 10$ which gives $q = 5$

Uses common point to find q

(M1A1)

Another point common to both is $(3 + p, 8, 0)$

Hence $\begin{pmatrix} 3+p \\ 8 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 10$ which gives $p = -1$

Use of second point and value of q to find p or

consideration of scalar product $\begin{pmatrix} p \\ 3 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$

(M1A1)

(4)

(c) (i) $\mathbf{n} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \quad \mathbf{d} = \begin{pmatrix} p \\ 3 \\ -1 \end{pmatrix}$

Let α be angle between normal and direction ratios
(plane) (line)

$$\mathbf{n} \cdot \mathbf{d} = p + 1$$

$\mathbf{n} \cdot \mathbf{d}$ correct

M1

$$\sin \theta = \frac{1}{\sqrt{6}} \Rightarrow \cos \alpha = \frac{\pm 1}{\sqrt{6}}$$

Correct $\cos \alpha$ stated or implied

B1

$$\Rightarrow \frac{p+1}{\sqrt{6}\sqrt{p^2+10}} = \frac{\pm 1}{\sqrt{6}}$$

Forming equation connecting all relevant parts **and**
attempting to solve for p (condone missing $\pm$)
Dependent on first M1 – fully correct for A1

m1A1

$$\Rightarrow (p+1)^2 = p^2 + 10$$

$$\Rightarrow p^2 + 2p + 1 = p^2 + 10$$

$$\Rightarrow 2p = 9 \text{ giving } p = 4.5$$

CAO

A1

5

Alternative:

$$|\mathbf{n} \times \mathbf{d}| = \sqrt{49 + (1+2p)^2 + (3-p)^2}$$

$|\mathbf{n} \times \mathbf{d}|$ correct

(M1)

$$\sin \theta = \frac{1}{\sqrt{6}} \Rightarrow \sin \alpha = \frac{\sqrt{5}}{\sqrt{6}}$$

Correct sin α stated or implied

(B1)

$$\frac{\sqrt{49 + (1+2p)^2 + (3-p)^2}}{\sqrt{6}\sqrt{p^2+10}} = \frac{\sqrt{5}}{\sqrt{6}}$$

Forming equation connecting all relevant parts **and**
attempting to solve for p . Dependent on first M1 –
fully correct for A1

(m1A1)

Leading to $p = 4.5$
CAO

(A1)

(5)

(ii) $z = 2 \Rightarrow t = -1 \Rightarrow x = -1.5$

$p = 4.5$ $y = q - 3$

$\Rightarrow -1.5 + q - 3 + 4 = 10$

Attempt to form an equation for q using $t = -1$

M1

$q = 10.5$
CAO

A1

EXAM PAPERS PRACTICE ² [12]

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Q3.

(a) Use of scalar product on

$\begin{pmatrix} 2 \\ 1 \\ 7 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix}$

M1

Sc.Prod. = ± 21 ;

B1

Moduli $\sqrt{54}$ and $\sqrt{26}$ correct
Accept 7.348... & 5.099...

B1

AWRT 56°

A1

- (b) $2x + y + 7t = 10$ and $3x + y - 4t = 7$ noted or used

M1

Eliminating (say) y to get x as a fn. of t

M1

$$x = 11t - 3$$

CAO

A1

Subst^g. back for y

M1

$$y = 16 - 29t$$

CAO

A1

$$\frac{x+3}{11} = \frac{y-16}{-29} = z \quad (=t)$$

B1 ft

6

- (c) Attempt at either $\begin{pmatrix} \lambda+20 \\ 9\lambda-1 \\ 4\lambda+7 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 7 \end{pmatrix} = 10$ or $\begin{pmatrix} \lambda+20 \\ 9\lambda-1 \\ 4\lambda+7 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix} = 7$.

M1

Solving either

$$40 + 2\lambda - 1 + 9\lambda + 49 + 28\lambda = 10 \text{ or}$$

$$60 + 3\lambda - 1 + 9\lambda - 28 - 16\lambda = 7$$

M1

$$\lambda_1 = -2$$

$$\lambda_2 = 6$$

A1A1

$$P = (18, -19, -1) \text{ and } Q = (26, 53, 31)$$

$$PQ = \sqrt{8^2 + 72^2 + 32^2} = \sqrt{6272} = 56\sqrt{2}$$

NB P, Q not required:

$$d = |\lambda_1 - \lambda_2| \times |\mathbf{i} + 9\mathbf{j} + 4\mathbf{k}| = 8 \times 7\sqrt{2} = 56\sqrt{2} \quad \mathbf{M1 \ A1}$$

M1A1

6

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Q4.

(a) $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 12 \\ 15 \\ 16 \end{bmatrix} = 12 + 15 - 16 = 11$ shown

B1

1

(b) eqn., or use, of line incorporating

p.v. $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ and d.v. $\begin{bmatrix} 12 \\ 15 \\ 16 \end{bmatrix}$

M1

$$\frac{x-1}{12} = \frac{y-1}{15} = \frac{z+1}{16}$$

A1

2

(c) Use of $\mathbf{r} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + \lambda \begin{bmatrix} 12 \\ 15 \\ 16 \end{bmatrix}$ with $\lambda = \pm 2$

M1

(25, 31, 31) and (-23, -29, -33)

A1 A1

3

(d) **Method 1**

$$PQ = 5$$

B1

$$\text{Area } \triangle PMN = \frac{1}{2} \times PQ \times MN = 250$$

(Since $Q = \text{midpt. } MN$)

M1 A1

3

Method 2

$$\overrightarrow{PM} = \begin{bmatrix} 20 \\ 30 \\ 35 \end{bmatrix}, \overrightarrow{PN} = \begin{bmatrix} -28 \\ -30 \\ -29 \end{bmatrix}$$

Attempted

(M1)

$$\overrightarrow{PM} \times \overrightarrow{PN} = (\pm) 20 \begin{bmatrix} 9 \\ -20 \\ 12 \end{bmatrix} \text{ attempted}$$

||gm. or Δ

(M1)

within an area formula
Area $\Delta PMN = 250$

CAO

(A1)

[9]

Q5.

(a) (i) $\mathbf{r} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix}$

Any correct vector line equation; B1 if one vector correct, or if both correct but equation not in correct form

B2,1

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2

(ii) $\mathbf{r} = \begin{bmatrix} 1 & 4 & -3 \\ 2 & -1 & 0 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1+2\lambda \\ 2+3\lambda \\ 3+6\lambda \end{bmatrix} = \begin{bmatrix} -4\lambda \\ \lambda \\ -\lambda \end{bmatrix}$

Attempt at multiplication

M1

At least one entry correct

A1

All three correct

A1

Clear and valid explanation that this is a line through O

E1

4

(b) (i)
$$\begin{bmatrix} 1 & 4 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} p \\ \frac{1}{2}p + k \end{bmatrix} = \begin{bmatrix} 3p + 4k \\ \frac{3}{2}p - k \end{bmatrix}$$

For LHS

B1

For RHS

M1A1

Answer satisfies $y = \frac{1}{2}x - 3k$

A1

4

(ii) Equal gradients, hence parallel

ft if previous answer is of the form $y = \frac{1}{2}x + c$

E1F

Distance = $|k - c| \cos \theta$ with $\tan \theta = \frac{1}{2}$

Allow incorrect value of c here

M1

$\dots = \frac{8k}{\sqrt{5}}$

Allow 3.58k

A1

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Q6.

(a) (i) $\mathbf{a} = \mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} + 2\mathbf{j} + 6\mathbf{k}$

B1

1

(ii) Equating for $\lambda: \frac{x-1}{3} = \frac{y-1}{2} = \frac{z-2}{6}$

M1A1

2

(b) (i)
$$\mathbf{n} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 3 & 2 \\ 1 & 1 & 3 \end{vmatrix} = 7\mathbf{i} - 10\mathbf{j} + \mathbf{k}$$

M1A1

$$d = \begin{bmatrix} 7 \\ 5 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 7 \\ -10 \\ 1 \end{bmatrix} = 0$$

M1

ft n

A1

4

(ii) $d = 0 \Rightarrow$ plane through / contains the origin

B1

1

(c) $\sin \theta / \cos \theta = \frac{\text{scalar product}}{\text{product of moduli}}$
Must be $3i + 2j + 6k$ and their n

M1

Numerator = $21 - 20 + 6 = 7$
ft correct (unsimplified)

B1

Denominator = $7\sqrt{150}$
ft both correct (unsimplified)

B1

$\theta = 4.7^\circ$

CAO

A1

4

[12]

Q7.

(a) $AB^2 = (5 - 3)^2 + (3 - -2)^2 + (0 - 1)^2$
use $\pm (\overrightarrow{OB} - \overrightarrow{OA})$ in sum of squares of components
allow one slip in difference

M1

$$AB = \sqrt{30}$$

accept 5.5 or better

A1

(b) $\begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} = 2 + 3 = 5$

$\pm \overrightarrow{AB} \cdot$ direction l evaluated
condone one component error

M1

5 or -5

A1

$$\cos \theta = \frac{5}{\sqrt{30}\sqrt{10}}$$

F on either of candidates' vectors

B1F

use $|a| |b| \cos \theta = a \cdot b$; values needed

M1

$$\theta = 73^\circ$$

CAO
(condone 73.2, 73.22 or 73.22...)

A1

5

(c) $\overrightarrow{AC} = \begin{bmatrix} 5+\lambda \\ 3 \\ -3\lambda \end{bmatrix} - \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2+\lambda \\ 5 \\ -1-3\lambda \end{bmatrix}$

for $\overrightarrow{OC} - \overrightarrow{OA}$ or $\overrightarrow{OA} - \overrightarrow{OC}$ with $\overrightarrow{OC}$ in

terms of λ
condone one component error

M1A1

$$(2 + \lambda)^2 + 5^2 + (-1 - 3\lambda)^2 = 30$$

m1

$$10\lambda^2 + 10\lambda = 0$$

$$(\lambda = 0 \text{ or } \lambda = -1)$$

A1

$$(\lambda = 0 \Rightarrow (5, 3, 0) \text{ is } B)$$

$$\lambda = -1 \Rightarrow C \text{ is } (4, 3, 3)$$

condone $\begin{bmatrix} 4 \\ 3 \\ 3 \end{bmatrix}$

A1

5

[12]

Q8.

- (a) Use of $\cos \theta = \frac{\text{scalar product}}{\text{product of moduli}}$
Must be $(3\mathbf{i} - 4\mathbf{j} + \mathbf{k})$ and $(2\mathbf{i} + 12\mathbf{j} - \mathbf{k})$

M1

Numerator = ± 43

B1

Denominator = $\sqrt{26} \cdot \sqrt{149}$

Dr. = $5.099... \times 3.742... = 0.6908...$

B1

$\theta = 46.3^\circ$

A1

4

- (b) $3x - 4y + z = 2$ and $2x + 12y - z = 38$

B1 B1

2

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- (c) $(3\mathbf{i} - 4\mathbf{j} + \mathbf{k}) \times (2\mathbf{i} + 12\mathbf{j} - \mathbf{k})$

M1

$= -8\mathbf{i} + 5\mathbf{j} + 44\mathbf{k}$

A1

p.v. of any point on line

M1

e.g. $(0, 5, 22), (8, 0, -22), (4, 2\frac{1}{2}, 0)$

A1

$\frac{x-x_c}{-8} = \frac{y-y_c}{5} = \frac{z-z_c}{44}$

ft

B1F

5

Or Adding $\Rightarrow 5x + 8y = 40$ (e.g.)
Eliminating one variable

(M1)

$\frac{x-8}{-8} = \frac{y}{5} = \lambda$ Or $\frac{x}{-8} = \frac{y-5}{5} = \mu$
Parametrisation attempted

(dM1)
 (A1)

$x = 8 - 8\lambda,$ $x = -8\mu$
 $y = 5\lambda$ $y = 5 + 5\mu$
Substⁿ. to find third variable

(M1)

$\Rightarrow z = 44\lambda - 22$ $\Rightarrow z = 44\mu + 22$

$$\frac{x-x_c}{-8} = \frac{y-y_c}{5} = \frac{z-z_c}{44}$$

(A1)

(5)

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Q9.

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Use of $\mathbf{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3-2\lambda \\ 1+6\lambda \\ -1+3\lambda \end{bmatrix}$

B1

Equating for λ : $\frac{x-3}{-2} = \frac{y-1}{6} = \frac{z+1}{3}$

M1

A1

[3]

Examiner reports

Q1.

The two transformations were often correctly described, although on occasions the plane $z = 0$ became 'an axis' or 'line'. A small minority of students reversed the order of multiplication of the two matrices. In part (c) an eigenvector approach rarely scored many marks due to full determinant expansion errors, incorrectly setting the determinant of $\mathbf{M}_3 = 0$ or not realising that 1 was an eigenvalue. The approach of $\mathbf{M}_3\mathbf{v} = \mathbf{v}$ was far more successful apart from minor manipulation errors.

Q2.

This topic of equations of lines and planes is not well understood by students. Surprisingly, part (a) was not always answered correctly with arguments concerning the scalar product often wrongly used. Part (b) was most often successful when two points on the line were used or if the scalar product of the normal to the plane and direction vector of the line was set equal to zero and solved to find p . Part (c)(i) proved very discriminating with students using both scalar and vector products in tandem but not using the correct trigonometric ratio and confusing sine and cosine.

Q3.

This was another high-scoring question for many candidates, especially since the three parts were essentially completely independent of each other's results. Part (a) was straightforward and routine, though rather a lot of candidates lost marks by failing to give the acute angle required, subtracting the correct one from 90° , mixing up sines and cosines in the working, missing the minus sign in front of the "4" in the second of the two given normal vectors, or making numerical slips in adding up the three squares in each of the moduli.

Q4.

Around a quarter of all candidates made attempts only at part (a), for just the one mark, and around half the candidates failed to proceed beyond part (b). Even amongst these, there was a large number of candidates who apparently didn't know the difference between vector and cartesian equations, with many others not happy to work with the given normal vector.

Of that half of the candidature that did proceed beyond part (b), most of them decided that "a distance of 50 units" meant that λ was ± 50 , rather than ± 2 .

Even in the final part of the question, despite being given the coordinates of P and Q and the information that the line PQ was the height of a right-angled triangle with base 100, only about five candidates (of 336) appreciated that they could work out the area of the triangle just as it was. Even more disappointing than this was that so few could employ alternative vector methods to get the correct answer, mostly due to not having had much success with parts (a) to (c).

Q5.

In part (a)(i), most candidates knew how to convert the given equations into a suitable vector equation, but some lost a mark by writing ' $L = \dots$ ' rather than ' $\mathbf{r} = \dots$ '. Part (a)(ii) was likewise answered well but often not perfectly, the usual fault here being a failure to show

clearly why the line must pass through the origin.

Part (b) was found more difficult than part (a), even though it was in a two-dimensional context. Much confusion arose from the use of the same letters x and y to denote the coordinates of a general point before and after the transformation. Many candidates were able to carry out an appropriate matrix multiplication but were unsuccessful in their attempts to form a linear equation for the image line. In part (b)(ii), most candidates earned the first mark by comparing the gradients of the two lines, even if the equation of the second line was incorrect, but very few indeed realised that the 'distance' between the two parallel lines referred to the perpendicular distance between them.

Q6.

This question was handled very confidently by most candidates. Few appreciated that the value $d = 0$ meant that part (b)'s plane contained the origin; many had a stab at some perpendicularity arrangement instead. In part (c) a significant minority of candidates found the complementary angle to the one required.

Q7.

In part (a) relatively few candidates went directly to the distance formula, with most finding the vector $\overrightarrow{AB}$ first. Some stopped there and thus gained no credit. Many candidates obtained the correct result, with any errors made usually occurring in misread signs or coefficients.

For part (b) many candidates identified the correct vectors to use and found the required angle correctly. The common error was to use the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ and OB ; such candidates gained partial credit for knowledge and use of the appropriate formula. Other candidates had a mix of vectors, and for some of these it was not possible to see what was intended and where they had got their vectors from. Some candidates used one pair of vectors for a scalar product and another pair for the moduli in their formula for $\cos \theta$.

There were few complete good answers to part (c), although some candidates were very inventive in their solutions, showing good geometrical insight. Many candidates showed little understanding of the geometry, with those who drew a diagram usually being the more successful. Many candidates equated the vectors $\overrightarrow{AC}$ and $\overrightarrow{AB}$ and went on to find B and C were the same point, without apparently realising any error had been made. Some candidates attempted to find the intersection of line l with line AB , again not noting the intersection point was B . Some candidates did take the expected approach and found $\overrightarrow{AC}$ in terms of parameter λ , but then didn't know what to do with it. Some who did equate the distances, made an error in their algebra, and ultimately few fully correct solutions were seen.

Q8.

This was the longest question on the paper, and contained a mixture of ideas from across the unit's topics. Most candidates coped very favourably with this variety and, minor arithmetical slips apart, marks were high. The biggest obstacle to success came in part (b)(i), where candidates needed to find the coordinates of any one point on the line of intersection. Setting x , y or z equal to zero and finding a value for the other two variables from the resulting simultaneous equations was the most common approach, and this worked well for most candidates.

Q9.

This was quite a straightforward test of some vector ideas. Most candidates were very happy to run through it with little difficulty.



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